Departement Elektriese, Elektroniese en Rekenaar-Ingenieurswese

Department of Electrical, Electronic and Computer Engineering

Eksamen

Examination

Module EBN111

Elektrisiteit en Elektronika

21 Junie 2011

Module EBN111

Electricity and Electronics

21 June 2011

UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Eksamensinligting: ***Exam information:***			
Maksimum punte: *Maximum marks:*	90	Volpunte: *Full marks:*	90
Duur van vraestel: *Duration of paper:*	180 minute *180 minutes*	Oopboek / toeboek: *Open / closed book:*	Toe *Closed*
Totale aantal bladsye (hierdie blad ingesluit): *Total number of pages (including this page):*		29	

BELANGRIK- ***IMPORTANT***

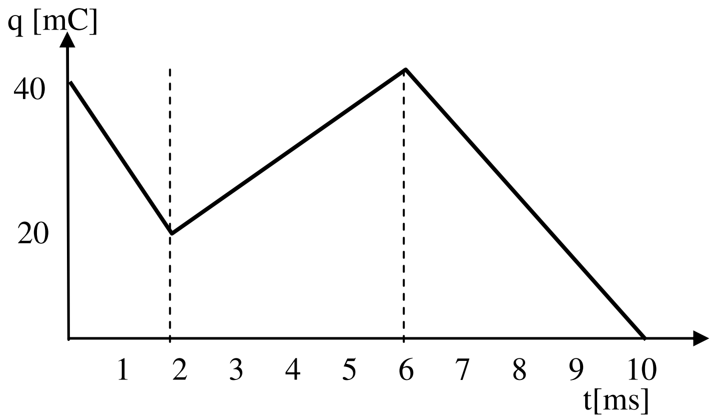
1. Die eksamenregulasies van die Universiteit van Pretoria geld.
   The examination regulations of the University of Pretoria apply.
2. Alle vrae moet op die ANTWOORDBLAD beantwoord word.
   All Questions must be answered on the ANSWER SHEET.
3. Studente mag hul eie rowwe berekeninge op die vraestel doen – die vraestel word nie ingeneem nie.
   Students may do their own rough calculations on the question paper – the question paper will not be taken in.
4. Polêre vorm: $A\angle\theta$; Reghoekige vorm: $a + jb$
   Polar form: $A\angle\theta$ *; Rectangular form:* $a + jb$

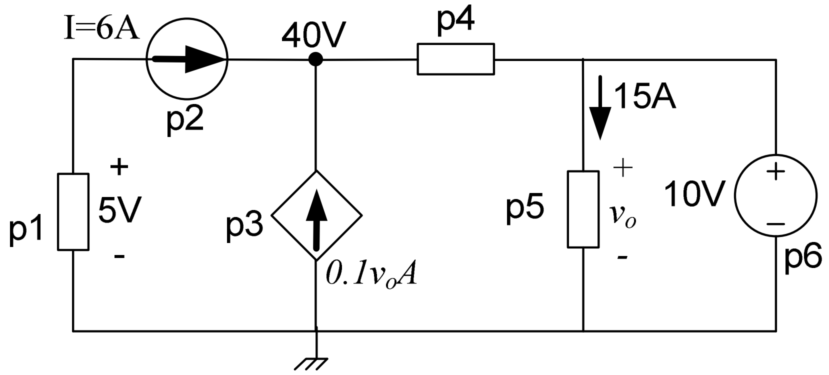
Dosent(e): *Lecturer(s):*	1 J Joubert 2 DV Bhatt

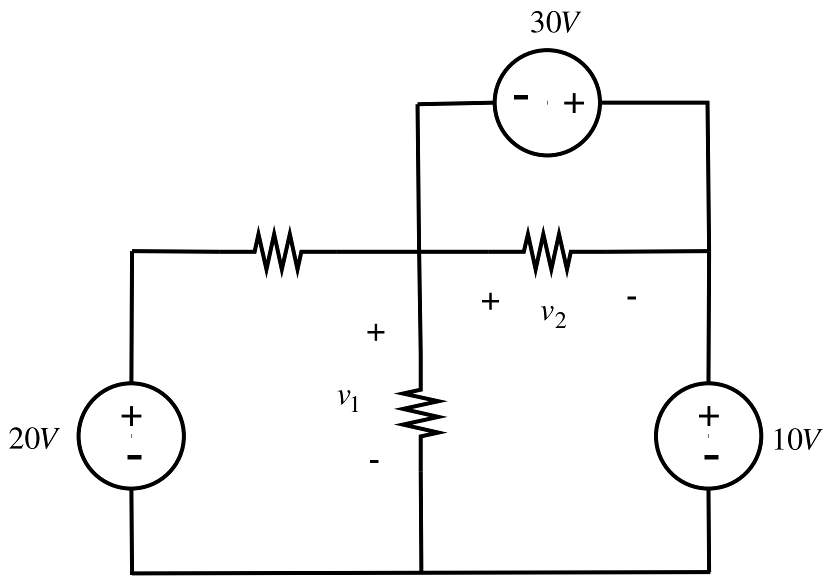
Groephoof: Ek bevestig dat die vraestel die uitkomstes toets soos gespesifiseer in die studiehandleiding.
Group head: I confirm that the question paper evaluates the outcomes as specified in the study guide.

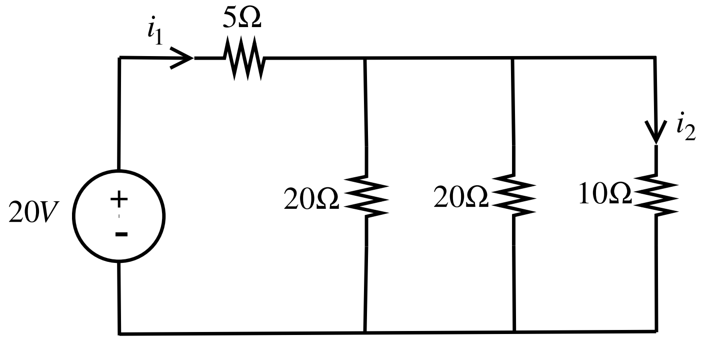
Groephoof: *Group Head:*	Prof J Joubert	Handtekening: *Signature:*	

VRAAG/QUESTION 1 [15]

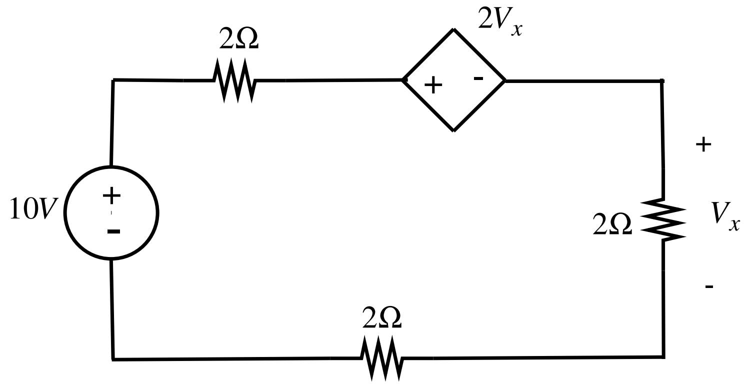
Vraag 1.1	**Question 1.1**
	
Lading wat deur ‘n element vloei is in die grafiek geplot. (a) Lei ‘n vergelyking af vir lading (in Coulomb) as funksie van tyd vir $2ms \le t \le 6ms$. [2] (b) Bereken die stroom by $t = 4$ ms. [1] (c) Bereken die totale lading wat teen $t = 2ms$ deur die element gevloei het. [1]	Charge flowing through an element is plotted in the graph. (a) Derive an expression for the charge (in Coulomb) as a function of time for $2ms \le t \le 6ms$. [2] (b) Calculate the current at $t = 4ms$. [1] (c) Calculate the total charge that has flowed through the element at $t = 2ms$. [1]

<table>
<tr><th>Vraag 1.2</th><th>Question 1.2</th></tr>
<tr><td colspan="2">
</td></tr>
<tr><td>Bepaal die drywing ge-assosieer met p3, p4 and p6. [3]</td><td>Determine the power associated with p3, p4 and p6. [3]</td></tr>
</table>

<table>
<tr><th>Vraag 1.3</th><th>Question 1.3</th></tr>
<tr><td colspan="2">
</td></tr>
<tr><td>Bepaal v_1 en v_2 in die stroombaan. [2]</td><td>Determine v_1 and v_2 in the circuit. [2]</td></tr>
</table>

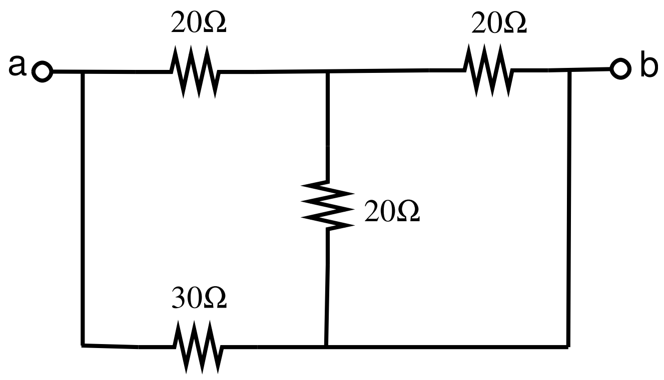
<table>
<tr><th>Vraag 1.4</th><th>Question 1.4</th></tr>
<tr><td colspan="2">
</td></tr>
<tr><td>Bepaal i_1 en i_2 in die stroombaan. [2]</td><td>Determine i_1 and i_2 in the circuit. [2]</td></tr>
</table>

Vraag 1.5	Question 1.5



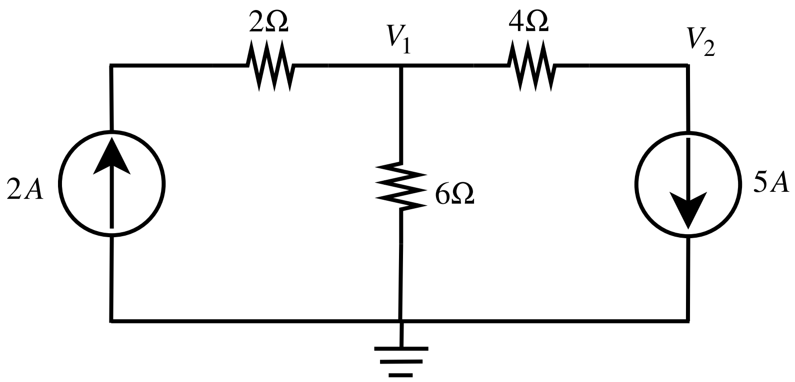
Bepaal V_x in die stroombaan. [2]	Determine V_x in the circuit. [2]

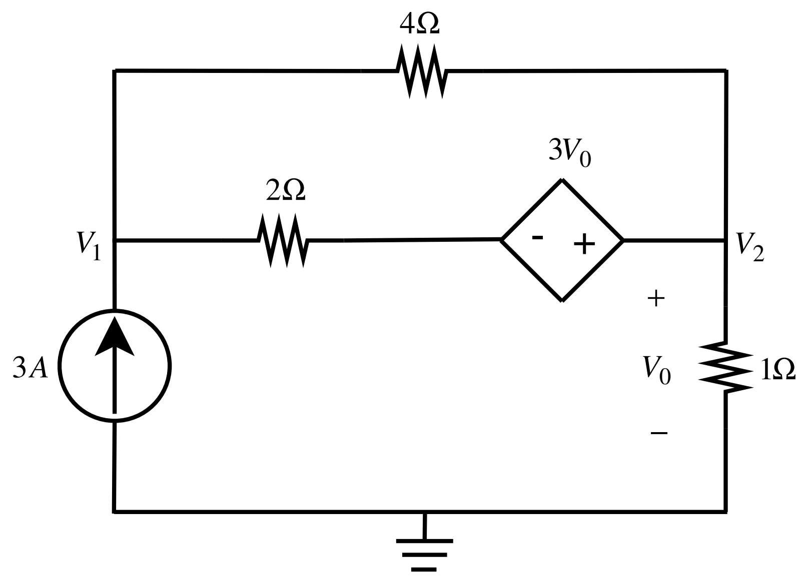
Vraag 1.6	**Question 1.6**

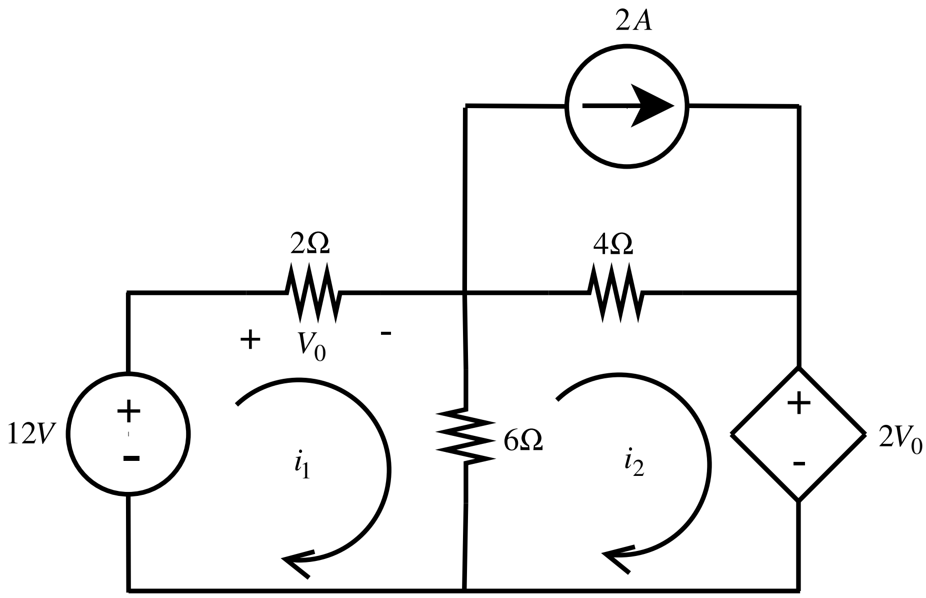


Reduseer na ′n enkele weerstand tussen terminale a-b. [2]	Reduce to a single resistor between terminals a-b. [2]

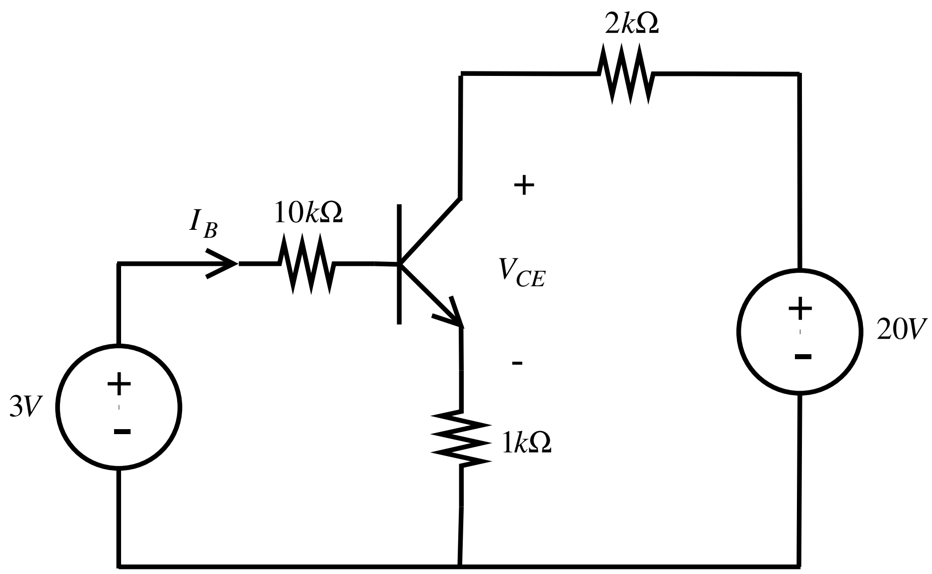
VRAAG/QUESTION 2 [40]

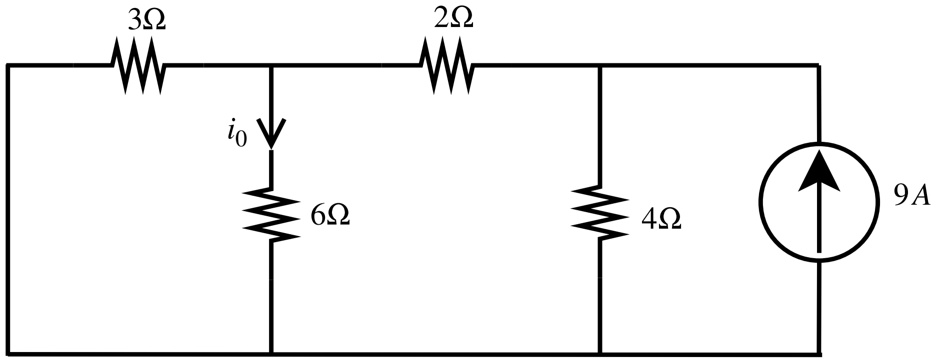
<table>
<tr><th>Vraag 2.1</th><th>Question 2.1</th></tr>
<tr><td colspan="2">
</td></tr>
<tr><td>Gebruik node-analise om twee vergelykings van die volgende vorm op te stel vir die stroombaan:
$$aV_1 + bV_2 = 24$$ $$cV_1 + dV_2 = 20$$ [4]</td><td>Use node analysis to set up two equations of the following form for the circuit:
$$aV_1 + bV_2 = 24$$ $$cV_1 + dV_2 = 20$$ [4]</td></tr>
</table>

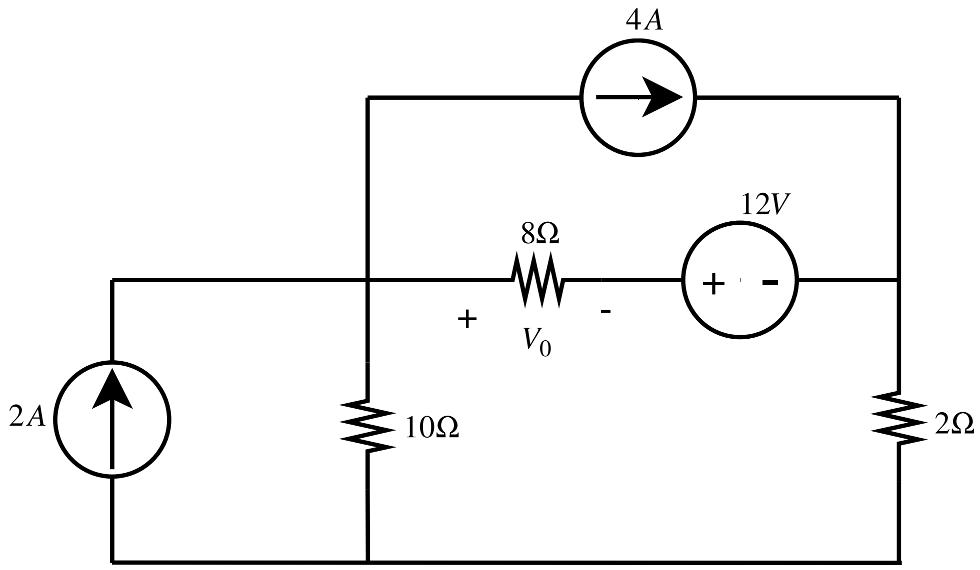
Vraag 2.2	Question 2.2
	
Gebruik node-analise om twee vergelykings van die volgende vorm op te stel vir die stroombaan: $$\begin{aligned} aV_1 + bV_2 &= 12 \\ cV_1 + dV_2 &= 0 \end{aligned}$$ [4]	Use node analysis to set up two equations of the following form for the circuit: $$\begin{aligned} aV_1 + bV_2 &= 12 \\ cV_1 + dV_2 &= 0 \end{aligned}$$ [4]

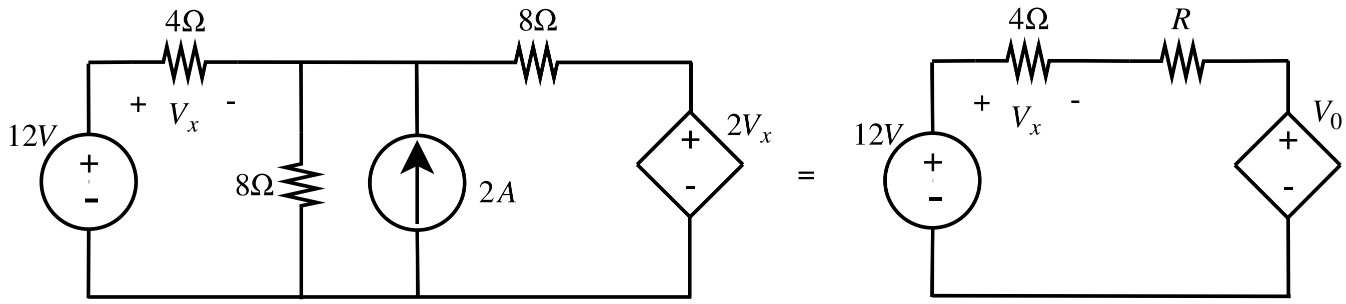
Vraag 2.3	Question 2.3
	
Gebruik lus-analise om twee vergelykings van die volgende vorm op te stel vir die stroombaan: $ai_1 + bi_2 = 12$ $ci_1 + di_2 = 8$ [4]	Use mesh analysis to set up two equations of the following form for the circuit: $ai_1 + bi_2 = 12$ $ci_1 + di_2 = 8$ [4]

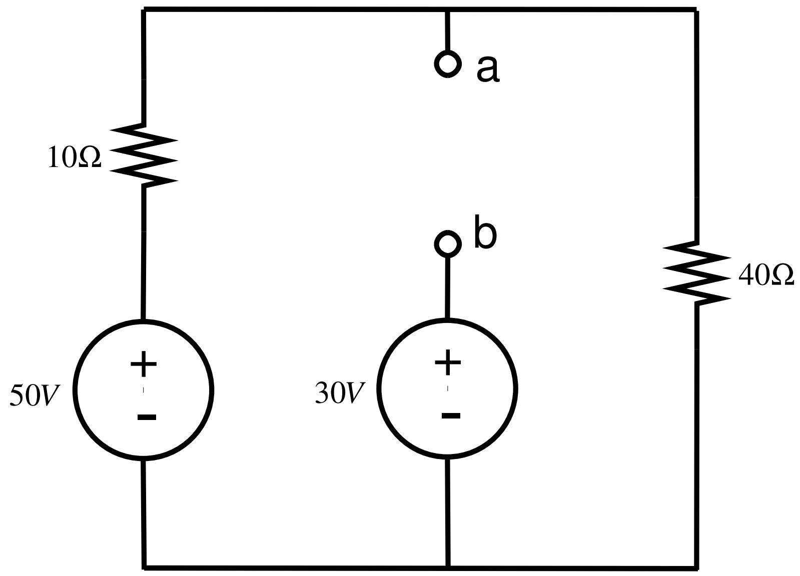
Vraag 2.4	Question 2.4
(a) Bepaal V_1. [2] $$\begin{bmatrix} -2 & 3 \\ 4 & 2 \end{bmatrix}\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$ (b) Bepaal i_3. [2] $$\begin{bmatrix} 2 & -3 & 0 \\ 4 & 0 & 6 \\ -2 & 1 & 4 \end{bmatrix}\begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 0 \\ 1 \end{bmatrix}$$	(a) Determine V_1. [2] $$\begin{bmatrix} -2 & 3 \\ 4 & 2 \end{bmatrix}\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$ (b) Determine i_3. [2] $$\begin{bmatrix} 2 & -3 & 0 \\ 4 & 0 & 6 \\ -2 & 1 & 4 \end{bmatrix}\begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 0 \\ 1 \end{bmatrix}$$

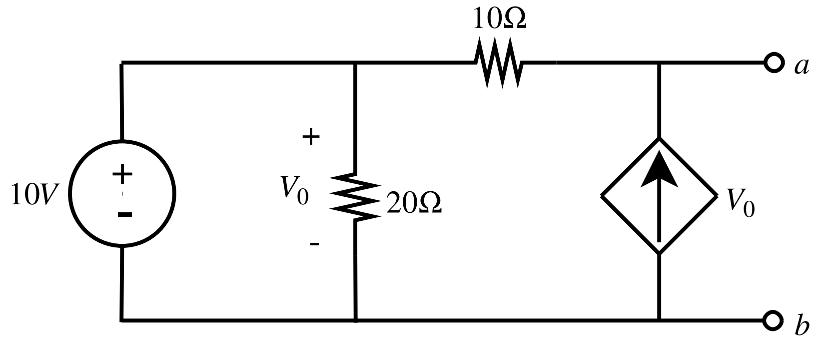
Vraag 2.5	Question 2.5
	
Bepaal I_B en V_{CE} in die stroombaan (aanvaar V_{BE} = 0.7V en β = 150). [4]	Determine I_B and V_{CE} in the circuit (assume V_{BE} = 0.7V and β = 150). [4]

Vraag 2.6	Question 2.6
	
(a) Gebruik lineariteit om i_0 te bereken. [2] (b) Met watter weerstand kan 'n mens die 6Ω weerstand vervang om te verseker dat optimum drywing in daardie weerstand gedissipeer sal word. [2]	(a) Use linearity to determine i_0. [2] (b) With what value resistor can one replace the 6Ω resistor to ensure that optimum power will be dissipated in that resistor. [2]

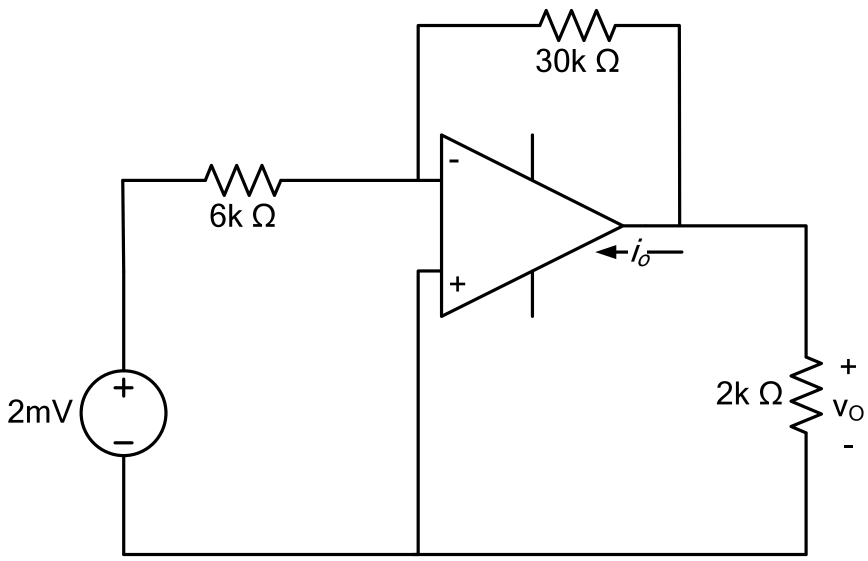
Vraag 2.7	Question 2.7
	
(a) Bepaal V_0 agv die *2A* bron. [2] (b) Bepaal V_0 agv die *12V* bron. [2]	(a) Determine V_0 due to the *2A* source. [2] (b) Determine V_0 due to the *12V* source. [2]

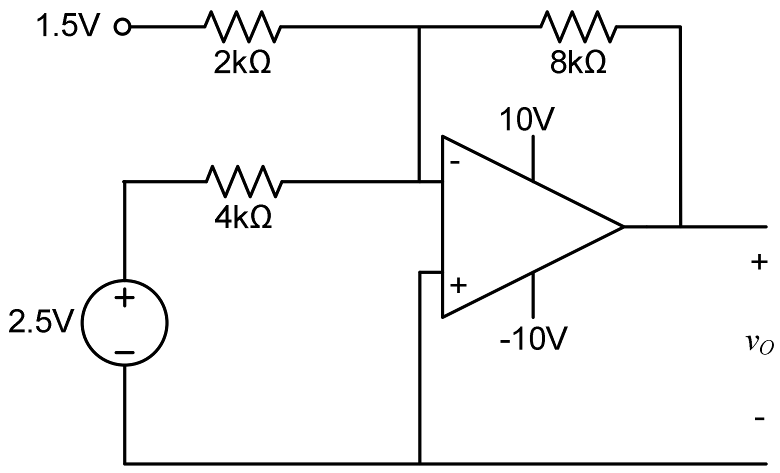
Vraag 2.8	Question 2.8
	
Gebruik brontransformasie om R en V_0 te bereken. [4]	Use source transformation to determine R and V_0. [4]

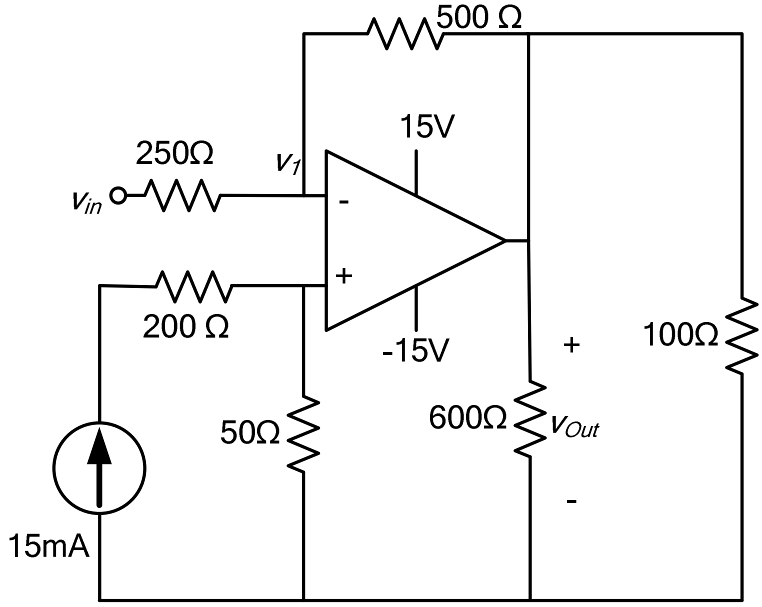
Vraag 2.9	2.9
	
Bereken die Thevenin ekwivalent tov terminale *a*-*b*. [4]	Determine the Thevenin equivalent wrt terminals *a*-*b*. [4]

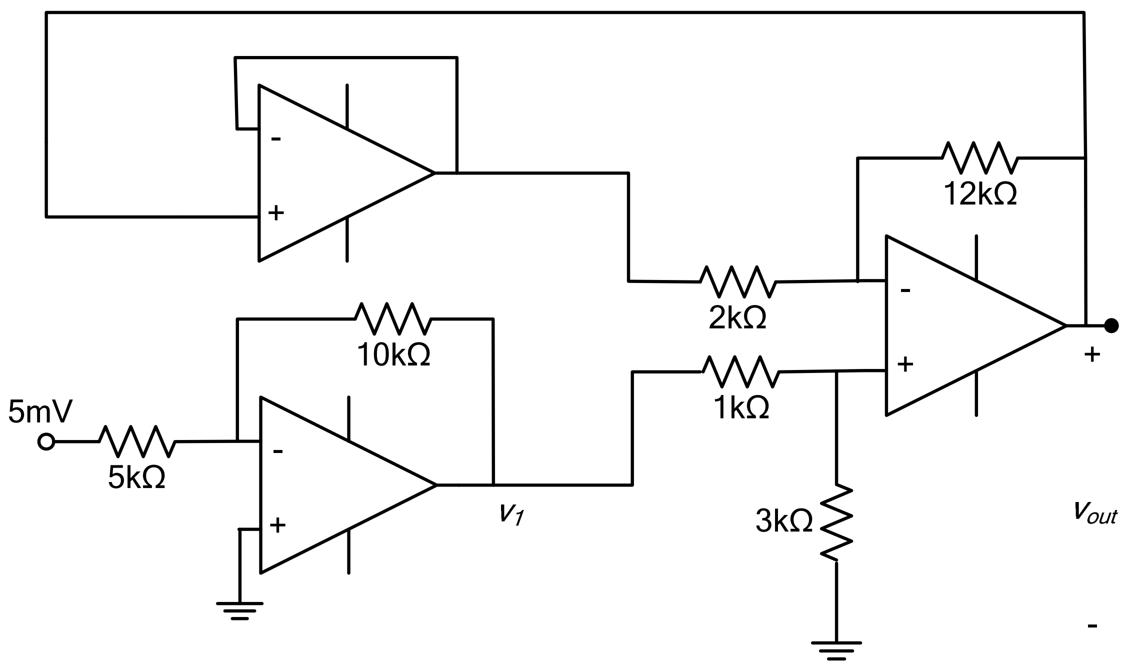
Vraag 2.10	Question 2.10
	
Bereken die Norton ekwivalent tov terminale *a*-*b*. [4]	Determine the Norton equivalent wrt terminals *a*-*b*. [4]

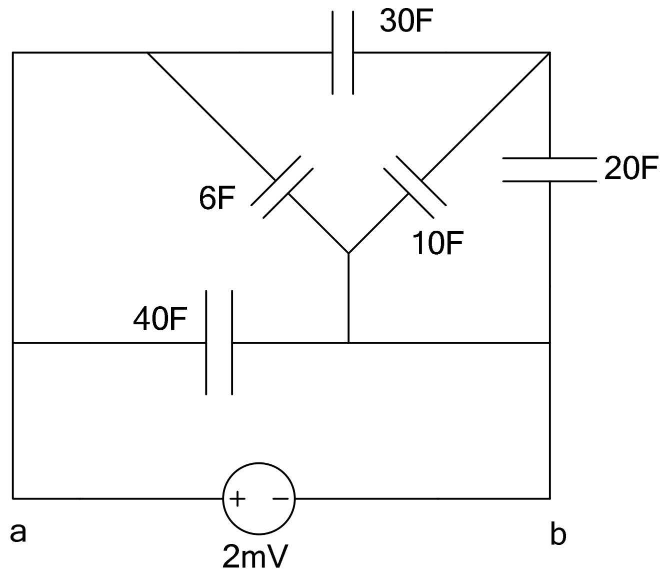
VRAAG 3 – QUESTION 3 [15]

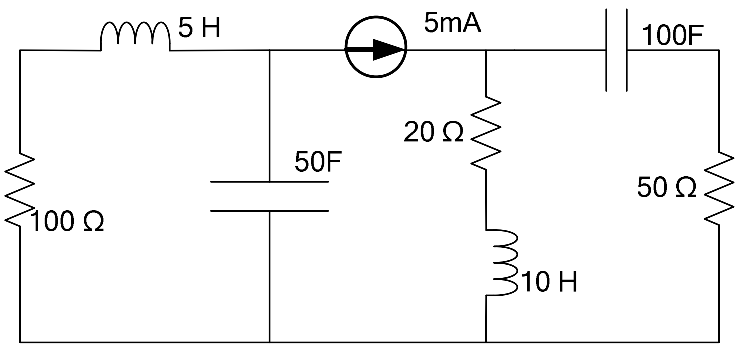
<table>
<tr><th>Vraag 3.1</th><th>Question 3.1</th></tr>
<tr><td colspan="2">
</td></tr>
<tr><td>Bepaal die waarde van i_o. [2]</td><td>Calculate the value of i_o. [2]</td></tr>
</table>

<table>
<tr><th>Vraag 3.2</th><th>Question 3.2</th></tr>
<tr><td colspan="2">
</td></tr>
<tr><td>Bepaal v_o. [2]</td><td>Calculate v_o. [2]</td></tr>
</table>

<table>
<tr><th>Vraag 3.3</th><th>Question 3.3</th></tr>
<tr><td colspan="2">
</td></tr>
<tr><td>Bepaal v_{in} if v_{out} = -10V. [2]</td><td>Determine v_{in} if v_{out} = -10V. [2]</td></tr>
</table>

Vraag 3.4	Question 3.4
	
Bepaal v_1 en v_{out}. [3]	Determine v_1 and v_{out}. [3]

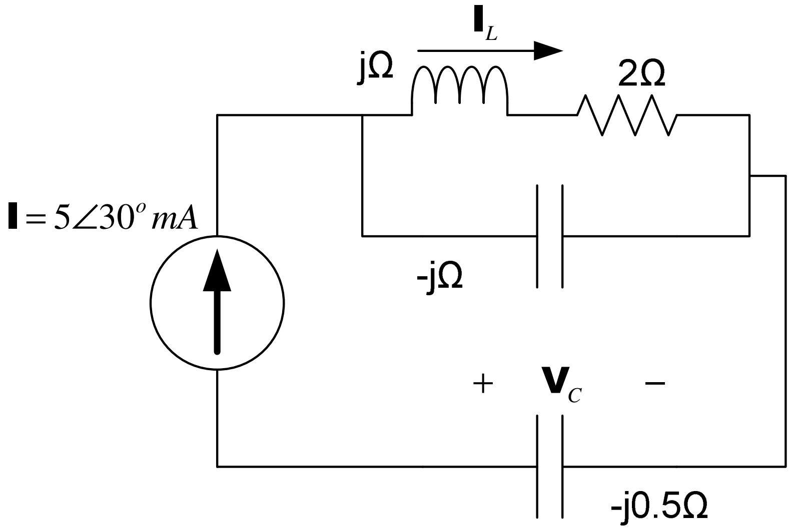
Vraag 3.5	Question 3.5
	
(a) Bereken die ekwivalente kapasitansie tussen terminale *a* en *b*. [2] (b) Bereken die energie gestoor in die 40F kapasitor. [1]	(a) Determine the equivalent capacitance between terminals *a* and *b*. [2] (b) Determine the energy stored in the 40F capacitor. [1]

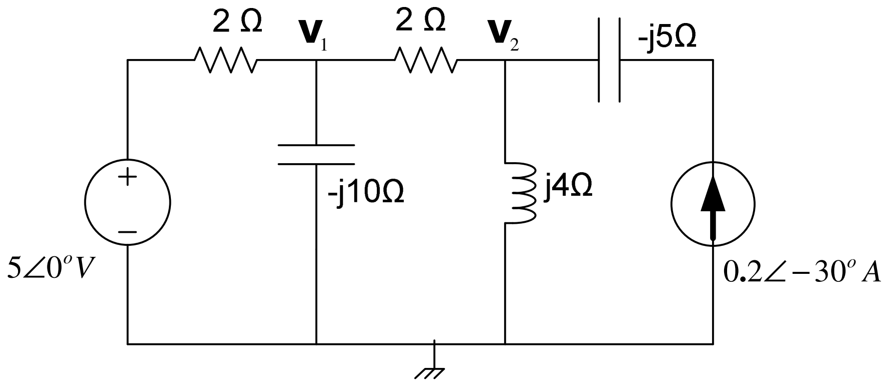
Vraag 3.6	**Question 3.6**
	
Bepaal energiewaardes in Joule: (a) gestoor in die 100F kapasitor, [1] (b) gestoor in die 10H induktor, en [1] (c) ge-assosieer met die 100Ω weerstand in een uur. [1]	Determine the energy in Joule: (a) stored in the 100F capacitor, [1] (b) stored in the 10H inductor, and [1] (c) associated with the 100Ω resistor in one hour. [1]

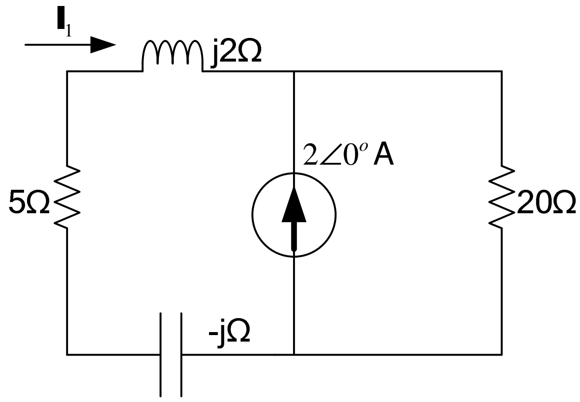
VRAAG 4 – QUESTION 4 [20]

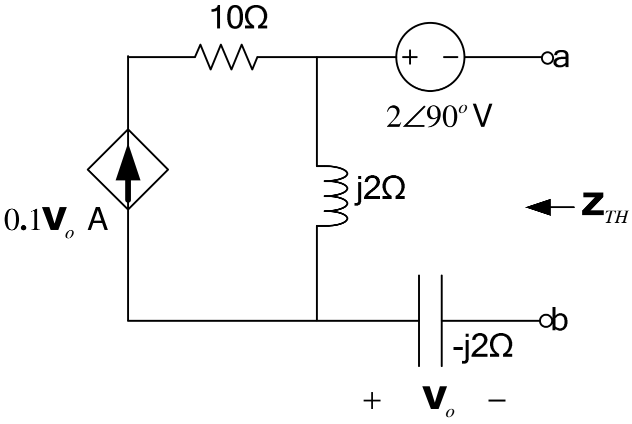
Vraag 4.1	Question 4.1
(a) Vereenvoudig en skryf in cosinus-vorm: $i(t)= 5\ sin(0.1t – 100^o) – cos(0.1t – 15^o)$ A. [2] (b) Watter van die volgende fasors loop voor, en met hoeveel grade? $v(t)=-6sin\ (10t+70^o)\ V$ en $i(t)=4cos\ (10t\text{-}45^o)$ A [2]	(a) Simplify and write the answer in cosine form: $i(t)= 5\ sin(0.1t – 100^o) – cos(0.1t – 15^o)$ A. [2] (b) Which of the following phasors is leading, and with how many degrees? $v(t)=-6sin\ (10t+70^o)\ V$ and $i(t)=4cos\ (10t\text{-}45^o)$ A [2]

Vraag 4.2	Question 4.2
(a) Evalueer die volgende (skryf die antwoord in polêre vorm): $$\mathbf{Z}_1 = \frac{e^{-j60^\circ} - 4}{(-1.01 + j1.5)}$$ [2] (b) Evalueer die volgende (skryf die antwoord in reghoekige vorm): $$\mathbf{Z}_2 = \frac{\left(3\angle -30^o\right)^* + 12.4 + j24.5}{\sqrt{25\angle -60^o}}$$ [2]	(a) Evaluate the following (write the answer in polar form): $$\mathbf{Z}_1 = \frac{e^{-j60^\circ} - 4}{(-1.01 + j1.5)}$$ [2] (b) Evaluate the following (write the answer in rectangular form): $$\mathbf{Z}_2 = \frac{\left(3\angle -30^o\right)^* + 12.4 + j24.5}{\sqrt{25\angle -60^o}}$$ [2]

<table>
<tr><th>Vraag 4.3</th><th>Question 4.3</th></tr>
<tr><td colspan="2">
</td></tr>
<tr><td>Bereken $\mathbf{V}_C$ en $\mathbf{I}_L$ (skryf antwoord in polêre vorm). [2]</td><td>Calculate $\mathbf{V}_C$ and $\mathbf{I}_L$ (write answer in polar form).[2]</td></tr>
</table>

<table>
<tr><th>Vraag 4.4</th><th>Question 4.4</th></tr>
<tr><td colspan="2">
</td></tr>
<tr><td>Gebruik node-analise om twee gelyktydige vergelykings vir fasors $\mathbf{V}_1$ en $\mathbf{V}_2$ op te stel. Skryf in die vorm
$$\alpha \mathbf{V}_1 + \beta \mathbf{V}_2 = 50$$
$$2\mathbf{V}_1 + \gamma \mathbf{V}_2 = \delta$$
met α, β, γ, and δ komplekse koeffisiënte in reghoekige vorm. [4]</td><td>Use nodal analysis to set up two simultaneous equations for phasors $\mathbf{V}_1$ and $\mathbf{V}_2$. Write in the form
$$\alpha \mathbf{V}_1 + \beta \mathbf{V}_2 = 50$$
$$2\mathbf{V}_1 + \gamma \mathbf{V}_2 = \delta$$
With α, β, γ, and δ complex coefficients in rectangular form. [4]</td></tr>
</table>

Vraag 4.5	Question 4.5
	
Bereken $\mathbf{I}_1$ mbv lus-analise. Skryf die finale antwoord in polêre vorm. [2]	Calculate $\mathbf{I}_1$ using mesh analysis. Write the final answer in polar form. [2]

Vraag 4.6	Question 4.6
	
Bereken die Thevenin ekwivalent $\mathbf{V}_{TH}$ en $\mathbf{Z}_{TH}$ tov terminale *a-b*. Skryf die antwoorde in reghoekige vorm. [4]	Calculate the Thevenin equivalent $\mathbf{V}_{TH}$ and $\mathbf{Z}_{TH}$ wrt terminals *a-b*. Write the answers in rectangular form. [4]

EINDE - END

EBN111 EKSAMEN ANTWOORDBLAD/ EXAM ANSWER SHEET
21 JUNIE/JUNE 2011

Student se besonderhede/*Student's details*			
Van en voorletters: *Surname and initials:*		Studentenommer: *Student number:*	
Graadkursus: *Degree Course:*			
Tel. nr.: *Tel no.:*			

PUNTE/MARKS									
Q1	**/15**	**Q2**	**/40**	**Q3**	**/15**	**Q4**	**/20**	**Final**	**/90**

VRAAG /QUESTION 1 [15]	
1.1	$i(t) =$
	$i(4ms) =$
	$q_{tot}(2ms) =$
1.2	$p3 =$
	$p4 =$
	$p6 =$
1.3	$v_1 =$
	$v_2 =$
1.4	$i_1 =$
	$i_2 =$
1.5	$V_x =$
1.6	$R_{eq} =$

VRAAG /QUESTION 2 [40]		
2.1	$a =$	$b =$
	$c =$	$d =$
2.2	$a =$	$b =$
	$c =$	$d =$
2.3	$a =$	$b =$
	$c =$	$d =$
2.4	$V_1 =$	$i_3 =$
2.5	$I_B =$	$V_{CE} =$
2.6	$i_0 =$	$R =$
2.7	$V_{0(2A)} =$	$V_{0(12V)} =$
2.8	$R =$	$V_0 =$
2.9	$R_{th} =$	$V_{th} =$
2.10	$R_N =$	$I_N =$

VRAAG 3 – QUESTION 3 [15]			
3.1	i_o =		
3.2	v_o =		
3.3	v_{in} =		
3.4	v_1 =	v_{out} =	
3.5	C_{eq} =	W_{40F} =	
3.6	W_{100F} =	W_{10H} =	$W_{100\Omega}$=

VRAAG 4 – QUESTION 4 [20]		
4.1	$i(t)$ =	
		Leading by:
4.2	$\mathbf{Z}_1$ =	
	$\mathbf{Z}_2$ =	
4.3	$\mathbf{V}_C$ =	
	$\mathbf{I}_L$ =	
4.4	α =	β =
	γ =	δ =
4.5	$\mathbf{I}_1$ =	
4.6	$\mathbf{V}_{TH}$ =	$\mathbf{Z}_{TH}$ =